A Universal Physics World Model for Autonomous Diagnosis and Correction of Computational Imaging

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Abstract

Computational imaging has transformed scientific discovery across scales, from sub-atomic microscopy to medical tomography. However, the reliability of these systems is often compromised by hidden "gate failures" in recoverability, stochastic noise, and forward-model mismatch. Current methods rely on manual parameter tuning or narrow, modality-specific calibration. Here we show a Physics World Model (PWM) that autonomously diagnoses and corrects imaging failures across 64 distinct modalities. By organizing physical knowledge into a "Triad Law"—balancing mathematical recoverability, quantum source limits, and system fidelity—PWM identifies the fundamental bottleneck in measured data and self-corrects the imaging operator. Our results demonstrate that PWM recovers high-fidelity information from previously "unsolvable" datasets in spectral imaging, electron microscopy, and magnetic resonance imaging. We anticipate that PWM will serve as a universal diagnostic instrument, enabling a new paradigm of self-aware, physically interpretable computational imaging.

1 Introduction

Computational imaging is fundamentally a decoding problem from physical carriers. Despite the success of deep learning, real-world translation is often bottlenecked by a lack of physical interpretability. We introduce the **Triad Law of Computational Imaging**, which posits that any reconstruction failure originates from one of three "gates":

- **Recoverability ($\mathcal{R}$):** The sufficiency of the sampling design relative to signal sparsity.

- **Stochasticity ($\mathcal{S}$):** The fundamental signal-to-noise ratio (SNR) limits imposed by physical carriers such as photons, electrons, and spins.
- **Fidelity ($\mathcal{F}$):** The accuracy of the forward-model operator, where sub-pixel mismatch can cause catastrophic failure.

2 Results

2.1 Cross-Modality Diagnostic Generality

PWM was validated across 64 modalities including optical microscopy and medical scanners. Utilizing three autonomous agents—the Compressed Agent, Source Agent, and Mismatch Agent—the system generates a “TriadReport” identifying specific physical causes of reconstruction artifacts.

2.2 Correcting the “Hidden Killer”: Operator Mismatch

The Mismatch Agent detected and corrected a 2-pixel mask shift in Coded Aperture Snapshot Spectral Imaging (CASSI), improving quality by $+4.8$ dB where standard neural networks failed. By fitting the forward-model parameters before final reconstruction, PWM transforms an imperfect measurement matrix into a physically corrected operator.

3 Discussion

PWM shifts the paradigm from physically informed models to physically interpretable world models. Unlike traditional black-box solvers, PWM’s diagnostic output allows scientists to understand why a measurement failed, distinguishing between the need for better algorithms, higher doses, or more precise hardware alignment. This transparency is critical for scientific discovery where the ”why” is as important as the result.

4 Methods

4.1 Formalization of the Triad Law

We define the computational imaging process as the recovery of a signal $x \in \mathbb{R}^N$ from measurements $y \in \mathbb{R}^M$. The recovery is governed by the objective function:

$$\hat{x} = \arg\min_x \mathcal{L}(y, A_\theta(x)) + \lambda\Phi(x)$$

where A_θ is the forward operator parameterized by θ, and $\Phi(x)$ is the prior.

4.2 The Source Agent: Carrier-Specific Noise Models

The Source Agent characterizes stochasticity $\mathcal{S}$ by selecting carrier-specific likelihoods:

- **Photons/Electrons (Poisson-Gaussian):** For shot-noise limited regimes:

$$y \sim \text{Poisson}(\alpha A_\theta x) + \mathcal{N}(0, \sigma^2)$$

- **Spins (MRI - Rician):** Models complex RF signal noise:

$$p(y|A_\theta x) = \frac{y}{\sigma^2} \exp\left(-\frac{y^2 + (A_\theta x)^2}{2\sigma^2}\right) I_0\left(\frac{yA_\theta x}{\sigma^2}\right)$$

4.3 The Mismatch Agent: Operator Correction Algorithms

The Mismatch Agent solves for optimal calibration parameters θ^* via reprojection error minimization:

$$\theta^* = \arg\min_{\theta \in \Omega} \|y - A_\theta(\hat{x}_\theta)\|_2^2 + \gamma\Gamma(\theta)$$

For discrete shifts, the agent utilizes a Beam Search algorithm, while continuous physics uses Gradient Descent:

$$\theta_{k+1} = \theta_k - \eta\nabla_\theta\|y - A_\theta(\hat{x})\|_2^2$$

4.4 The Compressed Agent: Recoverability Bounds

The agent computes a local Mutual Coherence $\mu(A)$ to verify if the sampling budget M satisfies the recovery condition:

$$M \geq c \cdot \mu^2(A) \cdot k \cdot \log(N)$$

Display Items

Figure 1: The PWM Diagnostic Paradigm. Panels showing (a) The "Three Gates" of the Triad Law, (b) The Physics Fidelity Ladder from Tier 0 to Tier 3, and (c) Comparison between failed standard reconstruction and PWM-corrected output.

Figure 2: Multi-Modality Validation. Performance metrics (PSNR/SSIM) and auto-generated TriadReports across 64 modalities.

Table 1: Benchmark Results. Quantitative comparison of the top 26 validated modalities against reference benchmarks.